

Time Allowed : 2 hrs 30 min

Total Marks : 300

Instructions

1. This Test Booklet contains **120** items (questions). Each item is printed in **English**. Each item comprises four responses (answers). You will select the response which you want to mark on the Answer Sheet. In case you feel that there is more than one correct response, mark the response which you consider the best. In any case, choose **ONLY ONE** response for each item.
2. You have to mark all your responses **ONLY** on the separate Answer Sheet provided. See directions in the Answer Sheet.
3. All items carry equal marks.
4. Before you proceed to mark in the Answer Sheet the response to the various items in the Test Booklet, you have to fill in some particulars in the Answer Sheet as per instructions.
5. **Penalty for wrong answers :**
THERE WILL BE PENALTY FOR WRONG ANSWERS MARKED BY A CANDIDATE IN THE OBJECTIVE TYPE QUESTION PAPERS.
 - (i). There are four alternatives for the answer to every question. For each question for which a wrong answer has been given by the candidate, **one-third** of the marks assigned to that question will be deducted as penalty.
 - (ii). If a candidate gives more than one answer, it will be treated as a **wrong answer** even if one of the given answers happens to be correct and there will be same penalty as above to that question.
 - (iii). If a question is left blank, i.e., no answer is given by the candidate, there will be **no penalty** for that question.

1. Let A and B matrices of order 3×3 . If $|A| = \frac{1}{2\sqrt{2}}$ and $|B| = \frac{1}{729}$, then what is the value of $|2B[\text{adj}(3A)]|$?
(a) 27 (b) $\frac{27}{2\sqrt{2}}$ (c) $\frac{27}{2}$ (d) 1
2. If z is any complex number and $iz^3 + z^2 - z + i = 0$, where $i = \sqrt{-1}$, then what is the value of $(|z| + 1)^2$?
(a) 1 (b) 4 (c) 81 (d) 121
3. What is the sum of all four digit numbers formed by using all digits 0, 1, 4, 5 without repetition of digits?
(a) 44440 (b) 46460 (c) 46440 (d) 64440
4. If x , y and z are the cube roots of unity, then what is the value of $xy + yz + zx$?
(a) 0 (b) 1 (c) 2 (d) 3
5. A man has 7 relatives (4 women and 3 men). His wife also has 7 relatives (3 women and 4 men). In how many ways can they invite 3 women and 3 men so that 3 of them are man's relatives and 3 of them are his wife's relatives?
(a) 340 (b) 484 (c) 485 (d) 469
6. A triangle PQR is such that 3 points lie on the side PQ , 4 points on QR and 5 points on RP respectively. Triangles are constructed using these points as vertices. What is the number of triangles so formed?
(a) 205 (b) 206 (c) 215 (d) 220
7. If $\log_b a = p$, $\log_d c = 2p$ and $\log_e f = 3p$, then what is $(ace)^{\frac{1}{p}}$ equal to?
(a) bd^2f^3 (b) bdf (c) b^3d^2f (d) $b^2d^2f^2$
8. If $-\sqrt{2}$ and $\sqrt{3}$ are roots of the equation $a_0 + a_1x + a_2x^2 + a_3x^3 + x^4 = 0$ where a_0, a_1, a_2, a_3 are integers, then which one of the following is correct?
(a) $a_2 = a_3 = 0$ (b) $a_2 = 0$ and $a_3 = -5$
(c) $a_0 = 6, a_3 = 0$ (d) $a_1 = 0$ and $a_2 = 5$
9. Let z_1 and z_2 be two complex numbers such that $\left| \frac{z_1 + z_2}{z_1 - z_2} \right| = 1$, then what is $\text{Re}\left(\frac{z_1}{z_2}\right) + 1$ equal to?
(a) -1 (b) 0 (c) 1 (d) 5
10. If $26! = n8^k$, where k and n are positive integers, then what is the maximum value of k ?
(a) 6 (b) 7 (c) 8 (d) 9
11. Consider the following statements in respect of two non-singular matrices A and B of the same order n :
 1. $\text{adj}(AB) = (\text{adj } A)(\text{adj } B)$
 2. $\text{adj}(AB) = \text{adj}(BA)$
 3. $(AB)\text{adj}(AB) - |AB|I_n$ is a null matrix of order nHow many of the above statements are correct?
(a) None
(b) Only one statement
(c) Only two statements
(d) All three statements